

The Relativistic Extension of the Active Vacuum Framework

Pramod Kumar Meher
C. V. Raman Global University, Bhubaneswar

June 2026

Abstract

The Active Vacuum Framework (AVF) treats the vacuum as an active energy distribution possessing a uniform Poincaré invariant ambient energy density $\mathcal{A}$. Matter produces a local depletion of this energy density, giving rise to a deformation field that accounts for the physical effects ordinarily attributed to gravitation. This paper develops the relativistic extension of the AVF through two more postulates: (i) first, the local vacuum response delay is inversely proportional to the available vacuum-energy density; (ii) second, the response delay mirrors the relativistic time-dilation behaviour established by Special Relativity (SR). These two postulates are used to define the general non-linear deformation law uniquely, which reproduces the temporal and spatial structure of the Schwarzschild solution exactly, with the horizon following as a consequence rather than being imposed. The framework's own requirement that a scalar cannot carry the deformation once the source is in motion is then made explicit and extended to a symmetric rank-two tensor field, of which the scalar law is the static, spherically symmetric component. That tensor is required as a structural necessity; it is not identified with the spacetime metric, and no field equation of general relativity is imposed upon it.

1 Introduction

The Active Vacuum Framework (AVF) treats the vacuum as a real, omnipresent energy distribution of density $\mathcal{A}$, identified with the measured dark-energy density, and treats matter as a localised confinement of that energy. In the companion paper that establishes the framework's foundations [1], five postulates fix the vacuum as omnipresent, matter as confined vacuum energy, the ambient background as uniform and Poincaré invariant, the vacuum as the sole active agent, and the gradient of the vacuum-energy deformation as the origin of force. From these, the Newtonian deformation profile $\mathcal{D}_G(r) = \mathcal{A}GM/(c^2r)$ and Newton's law of gravitation follow directly, with the deformation deepest near the source and decreasing as $1/r$.

Those five postulates are stated entirely in the language of static configurations: mass, spatial gradient, and the minimisation of total energy. None of them refers to motion, to

speed, or to the behaviour of the deformation field under a change of reference frame. They are complete for the static, weak-field regime, and they say nothing, by construction, about how the deformation profile is to be corrected once a body moves at an appreciable fraction of c and the slowing of local processes can no longer be neglected. This is not an oversight in the original postulates but a boundary of their scope: the variational principle they rest on is non-relativistic, and a non-relativistic principle cannot, on its own, single out the relativistic correction to the law it generates.

This paper develops the relativistic extension of the AVF through two more postulates: (i) first, a property of vacuum energy: the local vacuum response delay is inversely proportional to the available vacuum-energy density; (ii) second, the response delay behaviour mirrors the relativistic time-dilation behaviour established by Special Relativity (SR) [2, 3]. These postulates together relate the deformation profile at radius r to the Lorentz factor of a free-fall trajectory.

The framework's own internal requirement, that deformation in the relativistic regime cannot be represented by a single scalar once the source is in motion, is then made explicit and extended to a symmetric rank-2 tensor field, of which the scalar law derived here is the static, spherically symmetric component. It does not identify that tensor with the spacetime metric, and it imposes no field equation upon it.

2 Foundational Postulates of AVF

The AVF begins with five foundational postulates that define the nature of the vacuum, matter, and force.

Postulate 1 (The Vacuum Is Omnipresent). *The vacuum permeates all physical space at every scale. It is present wherever matter is not.*

Postulate 2 (Matter Is Localised Vacuum Energy). *Matter is a localised confinement of vacuum energy. The formation of any material object confines a portion of vacuum energy within a finite region of space.*

Postulate 3 (Ambient Vacuum Energy Is Uniform and Active). *Ambient vacuum energy is distributed uniformly throughout space with energy density $\mathcal{A}$. This ambient energy is dynamically active: it generates forces and governs the acceleration of material objects.*

Postulate 4 (The Vacuum Is the Sole Active Agent). *Only the vacuum generates force and produces acceleration. Material objects are passive. They do not directly exert forces on one another. All observed forces arise from the dynamics of the ambient vacuum energy acting on passive material objects.*

Postulate 5 (Deformation of Ambient Energy Generates Force). *Material entities locally deform the ambient vacuum-energy density. The force on a material object at any point is proportional to the local spatial gradient of the vacuum-energy deformation. Physical systems evolve toward configurations that minimise total energy.*

These five postulates are the foundation of AVF. The next section introduces a sixth and a seventh postulate that extend the framework to the relativistic regime.

3 The New Postulates

The ability of the vacuum to transmit signals, transport energy, and generate forces depends on the local vacuum-energy density. A reduction in the available vacuum-energy density therefore reduces the rate at which physical processes can occur and increases the characteristic response delay of the vacuum. In other words, the local rate of all physical processes is proportional to the available vacuum-energy density.

Consequently, regions of reduced vacuum-energy density exhibit a universal slowing of clocks, signal propagation, and dynamical processes relative to regions where the ambient vacuum-energy density remains undepleted.

Postulate 6 (Vacuum Response-Delay and Energy Density). *The normalized local vacuum-energy density, defined as the fraction of the ambient background $\mathcal{A}$ remaining available at radius r , is inversely related to the local vacuum response delay. Specifically:*

$$\frac{\mathcal{H}_G(r)}{\mathcal{A}} = \frac{\tau_0}{\tau(r)}, \quad (1)$$

where $\mathcal{A}$ is the undepleted ambient vacuum-energy density, $\mathcal{H}_G(r)$ is the local vacuum-energy density, τ_0 is the vacuum response delay in the undeformed region, and $\tau(r)$ is the local response delay.

Postulate 7 (Lorentz Kinematic Closure of Vacuum Response). *The local vacuum response delay is governed by the relativistic time-dilation factor associated with the kinematics of a freely falling observer. Specifically, the response delay scales as*

$$\tau(r) = \gamma(r) \tau_0, \quad (2)$$

where $\gamma(r)$ is the Lorentz time-dilation associated with the effective local velocity field of free fall as provided by SR [2, 3] local vacuum response delay is inversely proportional to the available vacuum-energy density; (ii) second, the response delay behaviour mirrors the relativistic time-dilation behaviour established by SR. These postulates together relate the deformation profile at radius r to the Lorentz factor of a .

This response delay at a point corresponds to a local process, occurring at r , which is expanded by a factor by which it runs slower relative to the same process if it runs in the undisturbed vacuum energy.

Remark 1. *The response delay increases by the Lorentz time-dilation factor by which all local physical processes occurring at radius r are slowed relative to the same process occurring far from any source of deformation. This results from the fact that vacuum depletion and time dilation are dual descriptions of the same underlying response asymmetry.*

Theorem 1 (Nonlinear Vacuum Deformation Law). *For a particle released from rest at spatial infinity and falling freely under the vacuum deformation field, the effective velocity at radius r is identified with the Newtonian escape velocity:*

$$v_{\text{esc}}^2 = \frac{2GM}{r}. \quad (3)$$

Consequently, the Lorentz factor is given by SR as

$$\gamma(r) = \frac{1}{\sqrt{1 - \frac{v_{\text{esc}}^2}{c^2}}} = \frac{1}{\sqrt{1 - \frac{2GM}{c^2 r}}}. \quad (4)$$

Combining Postulates 6 and 7 with Eq. (4), we obtain

$$\frac{\mathcal{A}}{\mathcal{H}_G(r)} = \gamma(r), \quad (5)$$

Using (4), we get

$$\frac{\mathcal{H}_G(r)}{\mathcal{A}} = \sqrt{1 - \frac{2GM}{c^2 r}} \quad (6)$$

We can write it as:

$$\frac{\mathcal{H}_G(r)}{\mathcal{A}} = \sqrt{1 - 2\mathcal{C}_G(r)}. \quad (7)$$

where the compactness parameter:

$$\mathcal{C}_G(r) = \frac{GM}{c^2 r} \quad (8)$$

3.1 Corollaries

Corollary 1 (The far-field boundary condition follows, rather than being imposed). As $r \rightarrow \infty$, $v_{\text{esc}}(r) \rightarrow 0$, so $\gamma(r) \rightarrow 1$ and $\mathcal{H}_G(r)/\mathcal{A} \rightarrow 1$. In the boundary-condition derivation of this law given elsewhere in this framework, $\mathcal{H}_G(r)/\mathcal{A} \rightarrow 1$ at infinity is imposed as a separate requirement; here it is not an additional assumption but a direct consequence of Postulate 6 itself, since a particle released from rest infinitely far away has, by definition, zero velocity there.

Corollary 2 (The horizon is where the escape velocity reaches c). The Newtonian escape velocity, Eq. (3), reaches exactly c at $r = 2GM/c^2$, the Schwarzschild radius derived in Section 3.2. Postulate 6 therefore gives the horizon a direct physical meaning, independent of the field-equation derivation given elsewhere in this framework: it is precisely the radius at which the classical, Newtonian formula for escape velocity would require a speed equal to that of light.

Corollary 3 (The interior is where the Newtonian formula already fails). For $r < 2GM/c^2$, Eq. (3) gives $v_{\text{esc}} > c$, so $\gamma(r)$, and correspondingly $\mathcal{H}_G(r)/\mathcal{A}$, become imaginary. This is the same imaginary result already recorded, without explanation, in the table of this framework's companion paper; Postulate 6 identifies its origin directly: it is the point at which the Newtonian escape-velocity formula itself, taken literally, would already demand faster-than-light motion, before any relativistic correction is applied to it at all.

Corollary 4 (Recovery of ordinary escape velocities in the weak-field limit). For $\mathcal{C}_G \ll 1$, $v_{\text{esc}}(r) = c\sqrt{2\mathcal{C}_G(r)}$ reduces to the ordinary Newtonian escape velocity used throughout classical mechanics. At the Earth's surface, where $\mathcal{C}_G \approx 7 \times 10^{-10}$, this gives $v_{\text{esc}} \approx 11.2$ km/s,

the Earth's known escape velocity, with $\gamma(r)$ indistinguishable from 1 to within one part in 10^9 . Postulate 6 therefore reduces smoothly to ordinary, non-relativistic experience wherever relativistic effects are negligible, with no separate weak-field postulate required.

Remark 2 (What this postulate does and does not assume). *The escape velocity, Eq. (3), is Newtonian. The Lorentz factor, Eq. (4), is special-relativistic. The identification in Postulate 6 is an independent assumption of this framework, not a consequence of general relativity. Unlike a derivation that identifies v_{esc} with the free-fall velocity of an observer in the Schwarzschild geometry specifically, justified by a coincidence that is itself a property of the Schwarzschild solution, the present derivation makes no reference to that solution or to any property established within general relativity. The exact match between Eq. (7) and the Schwarzschild result is, in this formulation, a consequence of the postulate, not an imported fact.*

3.2 Relativistic Correspondence

Equation (7) links directly to relativistic formulations as:

Time Dilation

$$\frac{dt_{\text{local}}}{dt_{\infty}} = \frac{\mathcal{H}_G(r)}{\mathcal{A}} = \sqrt{1 - 2\mathcal{C}_G(r)}. \quad (9)$$

Spatial Metric Stretching

$$\frac{ds_{\text{prop}}}{dr} = \frac{\mathcal{A}}{\mathcal{H}_G(r)} = \frac{1}{\sqrt{1 - 2\mathcal{C}_G(r)}}. \quad (10)$$

Gravitational Redshift

$$\frac{\nu_{\infty}}{\nu_{\text{loc}}} = \sqrt{1 - 2\mathcal{C}_G(r)}. \quad (11)$$

Remark 3 (A single origin for two effects). *Time dilation and spatial stretching are not two independently assumed relations. Both follow from the same function, $\mathcal{H}_G(r)/\mathcal{A}$, appearing directly in the temporal role and reciprocally in the radial role, the same structural pattern by which SR unifies time dilation and length contraction through a single invariant interval. The reciprocal structure, $ds_{\text{prop}}/dr = \mathcal{A}/\mathcal{H}_G(r)$, is itself a further structural assumption, consistent with but not derived from Postulate 6 alone.*

The Horizon Condition The horizon is the surface where the local vacuum-energy density vanishes, $\mathcal{H}_G(r_S) = 0$. From Eq. (7),

$$\sqrt{1 - \frac{2GM}{c^2 r_S}} = 0, \quad (12)$$

This implies:

$$r_S = \frac{2GM}{c^2}. \quad (13)$$

Weak-Field Limit For $\mathcal{C}_G \ll 1$,

$$\sqrt{1 - 2\mathcal{C}_G} \approx 1 - \mathcal{C}_G, \quad (14)$$

recovering the Newtonian deformation law, $\mathcal{H}_G(r) \approx \mathcal{A}(1 - \mathcal{C}_G(r))$, already established from the original five postulates.

4 Structural Axioms for Full Relativistic Extension

The five basis postulates of AVF provide the linear deformation law for a single static, spherically symmetric source. Two things separate that result from a full relativistic theory: the source may move, and the field must respond to changes causally rather than instantaneously. The presence of several sources at once is already handled by the linear superposition of the foundational deformation law. The two new postulates extend the linear deformation law to a non-linear relativistic regime. We need two more axioms, as stated below, to provide exactly the structure needed for the full relativistic extension of AVF.

Axiom 1 (Lorentz Covariance). *The laws governing vacuum deformation are invariant under local Lorentz transformations, and changes in the deformation field propagate causally at the speed of light c . This is the covariance requirement of SR. No requirement of invariance under general coordinate transformations, and no commitment to a dynamical, curved spacetime, is made here.*

Basis. SR is independently confirmed physics; a framework whose laws are not invariant under Lorentz transformations would predict different physics for observers in uniform relative motion, contrary to every relevant experimental test.

Implication. This axiom is what licenses the appearance of the Lorentz factor $\gamma(r)$ in Postulate 6 at all: without a prior commitment to Lorentz covariance, there would be no reason to expect a relativistic correction built from γ , rather than a Galilean correction, or no velocity-dependent correction at all, to be the relevant one. Axiom 1 is therefore the structural justification for the specific form Postulate 6 takes, not an independent, unrelated requirement.

Relevance to the relativistic extension. This axiom is the engine of the entire extension, and it does more than license the Lorentz factor already used. Applied to the static, spherically symmetric solution, Lorentz covariance fixes the field of a moving source from the field of a stationary one: the field seen when the source moves is obtained by boosting the static result. That boosted field is no longer spherically symmetric. It is compressed in the directions transverse to the motion and acquires components that depend on the velocity of the source, in the same way that the electric field of a moving charge does. Lorentz covariance therefore does not merely permit a relativistic correction to be made; applied to the static scalar result, it actively generates directional, motion-dependent structure that no scalar function of position can hold. This is the precise point at which the scalar description fails, and it is what makes the tensor of Axiom 2 unavoidable rather than optional.

Axiom 2 (Tensorial Structure of the Relativistic Deformation). *In the relativistic regime, a single scalar field is insufficient to describe vacuum deformation: the deformation must,*

in general, depend on direction relative to the source's motion, requiring a symmetric rank-2 tensor field $\mathcal{D}_{G,\mu\nu}(\mathbf{r},t)$, of which the scalar $\mathcal{D}_G(r)$ of Eq. (7) is the static, spherically symmetric, source-rest-frame component.

Basis. A scalar field can depend only on distance from its source, not on direction. Once the source is in motion, or once the deformation is examined by an observer in relative motion, the deformation it produces is no longer isotropic: it is stronger ahead of the motion than behind it, a directional asymmetry no scalar field can represent, in direct analogy with the boosted and gravitomagnetic fields of a moving mass already recognised as real effects.

Implication. This rules out, in principle, any attempt to extend Postulate 6 to a moving source using only the scalar $\mathcal{H}_G(r)/\mathcal{A}$ evaluated at a displaced position; the genuinely relativistic deformation of a moving source requires components beyond what a scalar field can carry. It does not, by itself, specify what those components are, or what equation governs them, only that scalar deformation cannot be the whole story once motion is admitted.

Relevance to the relativistic extension. This axiom records the direct consequence of the preceding one. Because boosting the static field produces anisotropy and velocity-dependent components, the relativistic deformation must be carried by an object capable of representing both direction and motion, and the minimal such object is a symmetric rank-2 tensor whose static, spherically symmetric, rest-frame component is the scalar law derived above. The gravitomagnetic field of a moving mass, a real and measured effect, is the concrete physical instance of structure that a scalar potential cannot describe and a tensor can. What this axiom deliberately does not do is fix the tensor's dynamics. It establishes that a tensor is required and identifies the scalar law as one of its components, while leaving entirely open what equation governs that tensor, and in particular whether that equation is, or is not, the one of general relativity. That open question, and the restraint behind it, are the subject of the closing remark.

Remark 4 (What is deliberately not assumed). *Axiom 2 requires that the relativistic deformation be tensor-valued; it does not identify this tensor with the spacetime metric $g_{\mu\nu}$, and no equation of the form $R_{\mu\nu} = 0$, $G_{\mu\nu} = 8\pi GT_{\mu\nu}/c^4$, or any other relation specific to general relativity is imposed on it anywhere in this paper. The tensorial requirement is a structural necessity, independent of whether that tensor turns out to coincide with, or differ from, the metric of general relativity; that question is left open.*

5 Conclusion

The relativistic extension of the Active Vacuum Framework requires two new postulates beyond the original five. The vacuum's response delay at a point is the relativistic time-dilation factor of a free-fall trajectory there, and the local vacuum-energy fraction is its inverse. Built from Newtonian energy conservation and SR alone, these postulates fix the nonlinear deformation law uniquely, with no free exponent, and reproduce the temporal and spatial structure of the Schwarzschild solution, with the horizon following as a consequence rather than an imported result. Time dilation and spatial stretching emerge as two aspects of the same function.

Extending the postulate beyond the static, spherically symmetric case requires the deformation to be tensor-valued. This is a structural necessity, stated here without identifying that tensor with the spacetime metric and without imposing the Einstein field equations upon it. What its governing equation is, if not that of general relativity, remains open.

References

- [1] P. K. Meher, *The Active Vacuum Framework: Fundamental Postulates and Physical Variables*, SSRN (2026).
- [2] R. P. Feynman, R. B. Leighton, and M. Sands, *The Feynman Lectures on Physics*, vol. I, Addison-Wesley, 1963.
- [3] R. Resnick, *Introduction to Special Relativity*, John Wiley & Sons, 1968.